Modern Database Systems

MASTER DIGITAL SCIENCE SOSE 2024

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Intro and Use Case

Universities should use data insights to enhance graduation rates and enriche the educational experience.

- Identify students in specific courses.
- Track student interactions with learning materials.
- Correlate interactions with grades.
- Analyze data to flag at-risk students and recommend personalized learning materials.
- Calculate average clicks and compare individual activities.
- Evaluate shared courses among students.

DB

PostgreSQL

- Offline, local

Neo4j AuraDB

- online, AuraDB free instance

Aim

Improve learning outcomes and prevent drop-outs.

Database Selection

PostgreSQL/ Relational Databases:

- PostgreSQL excels at structured data and ACID compliance for transactional applications.

Graph Databases (Neo4j):

- Ideal for university data with interconnected students, courses, grades, and materials.
- **Optimized Queries:** Neo4j simplifies complex queries like identifying common courses and tracking interactions.
- **Flexibility:** Neo4j adapts easily to new data types and relationships, crucial for dynamic educational environments.

Data

File name	Size
courses.csv	250 B
vle.csv	161 KB
assessments.csv	6 KB
studentInfo.csv	456 KB
studentAssessment.csv	2.2 MB
studentVle.csv	208.6 MB

Kaggle OULAD:

- Anonymized data on courses, students, and VLE interactions.

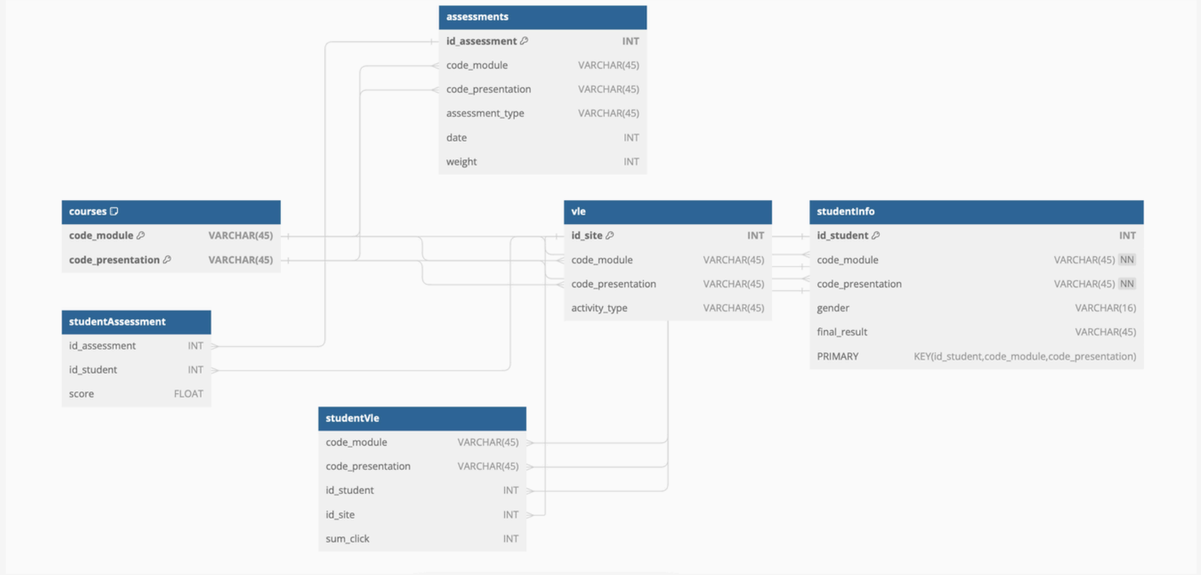
Data features

- Records of student enrolment, learning material usage, assessment performance.
- Facilitates research on online learning behavior

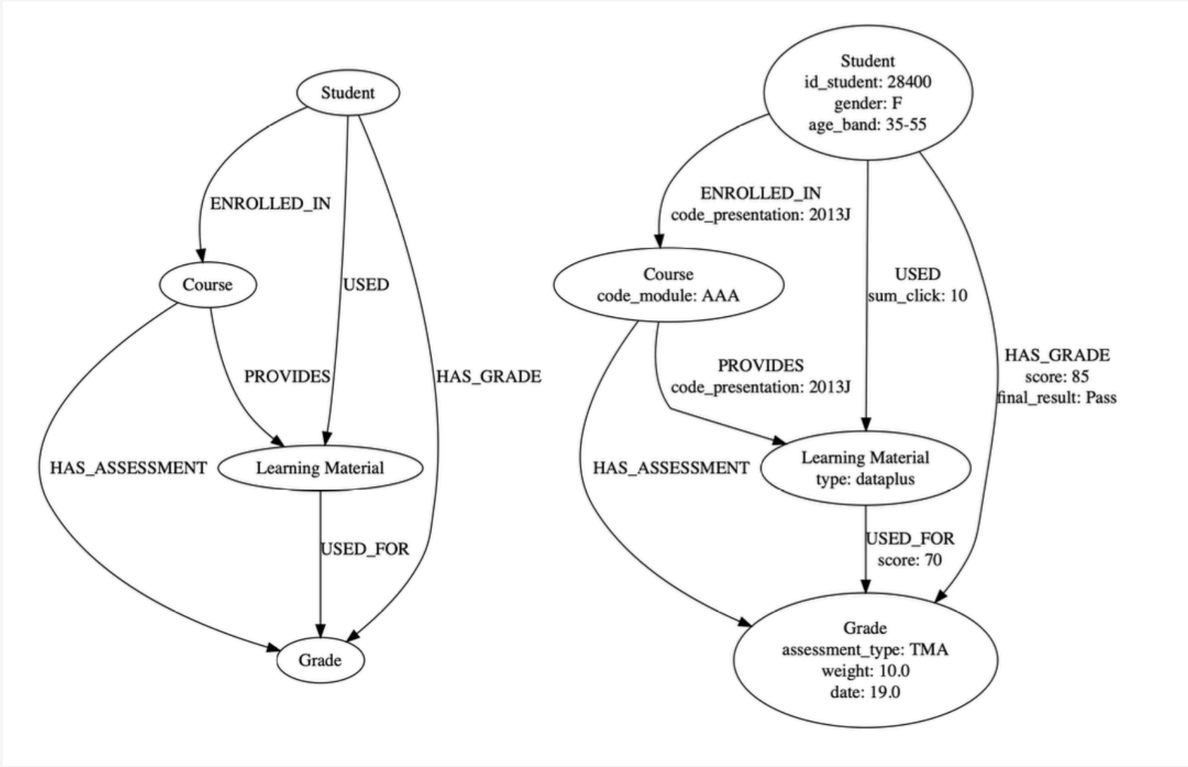
Structure

- Dataset stored in CSV files, connected through unique identifiers.
- Initial dataset size reduced for NoSQL database compatibility

Data Models



PostgreSQL



Neo4j

Queries

- Focus on these queries; may need additional queries for data deletion or updates.

Writing Query 1:
Inserts and updates student, course, material, and assessment records each term.

Reading Query 1:
Aggregates learning material clicks to analyze student engagement and performance correlations. Recommends materials clicked by former top students.

Reading Query 2:
The goal is to provide early alerts for students at risk by identifying those with lower than average engagement metrics.

Reading Query 3:
Connects students sharing common courses to recommend study partners.

Queries

Reading Query 4:
Lists students in a course for analyzing engagement and performance.

Reading Query 1: Recommendation of Learning Materials Based on Top Students

```sql
WITH QualifiedStudents AS (
    SELECT si.id_student
    FROM studentInfo si
    JOIN studentAssessment sa ON si.id_student = sa.
          id_student
    JOIN courses c ON si.code_module = c.code_module AND si.
          code_presentation = c.code_presentation
    WHERE c.code_module = 'AAA' AND LEFT(si.
          code_presentation, 4)::INT > 2013 AND sa.score >
          50.0
),
MostUsedMaterial AS (
    SELECT sv.id_student, v.activity_type, SUM(sv.sum_click)
           AS total_clicks
    FROM studentVle sv
    JOIN vle v ON sv.id_site = v.id_site
    WHERE sv.id_student IN (SELECT id_student FROM
          QualifiedStudents)
    GROUP BY sv.id_student, v.activity_type
    ORDER BY total_clicks DESC
)
SELECT activity_type, sum(total_clicks) AS clicks_total
FROM MostUsedMaterial
GROUP BY activity_type
ORDER BY clicks_total DESC;
```

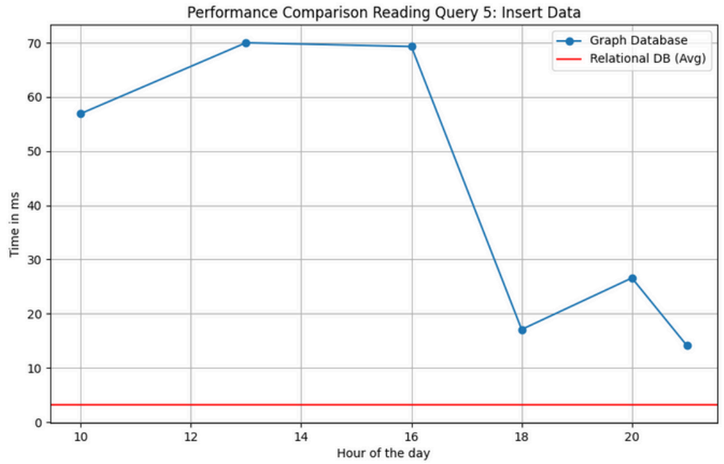
Listing 3: SQL Query

```cypher
// First MATCH clause to filter relevant courses and
      students
MATCH (c:Course {code_module: "AAA"})
WITH c

// Second MATCH clause to filter relevant ENROLLED_IN
      relationships
MATCH (s:Student)-[e:ENROLLED_IN]->(c)
WHERE toInteger(substring(e.code_presentation, 0, 4)) > 2013
WITH s, c

// Third MATCH clause to filter relevant HAS_GRADE
      relationships
MATCH (s)-[r:HAS_GRADE]->(g:Grade)-[:HAS_ASSESSMENT]-(c)
WHERE r.score > 50.0
WITH COLLECT(DISTINCT s) AS students

// Unwind to process the collected students further
UNWIND students AS s

// Fourth MATCH clause to process USED relationships and
      aggregate clicks
MATCH (s)-[u:USED]->(m:Learning_Material)
RETURN m.type as materialType, sum(u.sum_click) as
      total_clicks
ORDER BY total_clicks DESC;
```

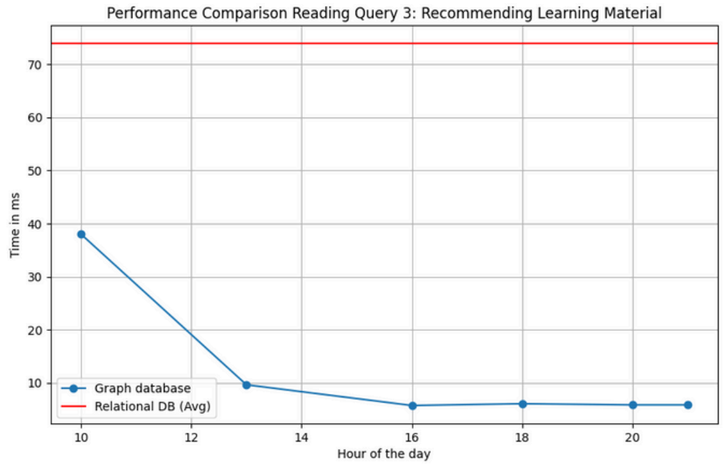
Listing 4: Cypher Query

Results

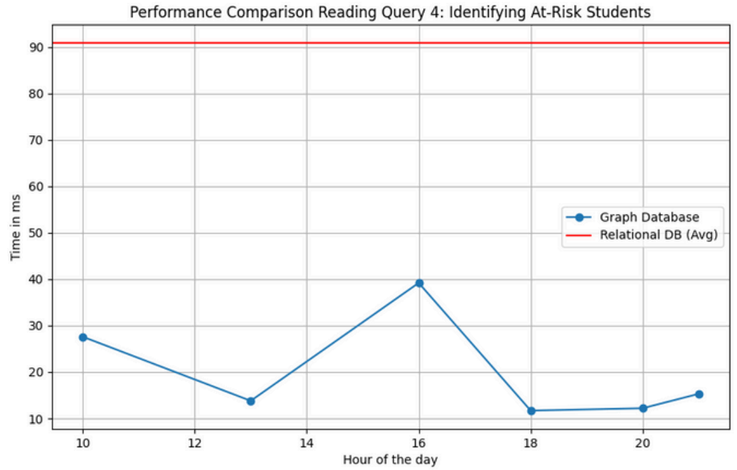
Writing Query



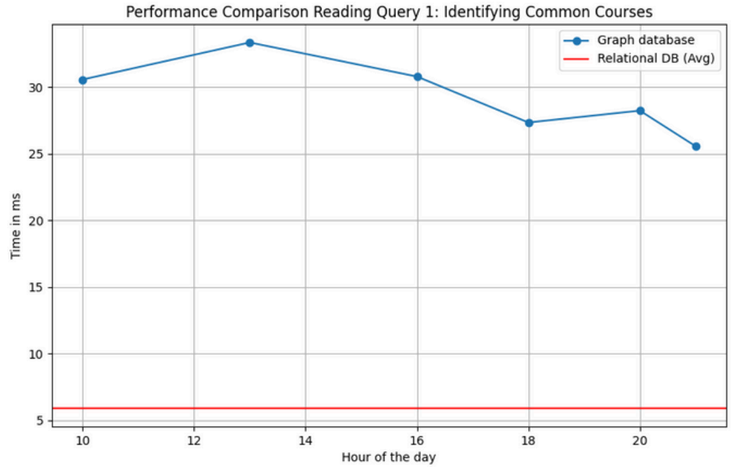
Recommendations



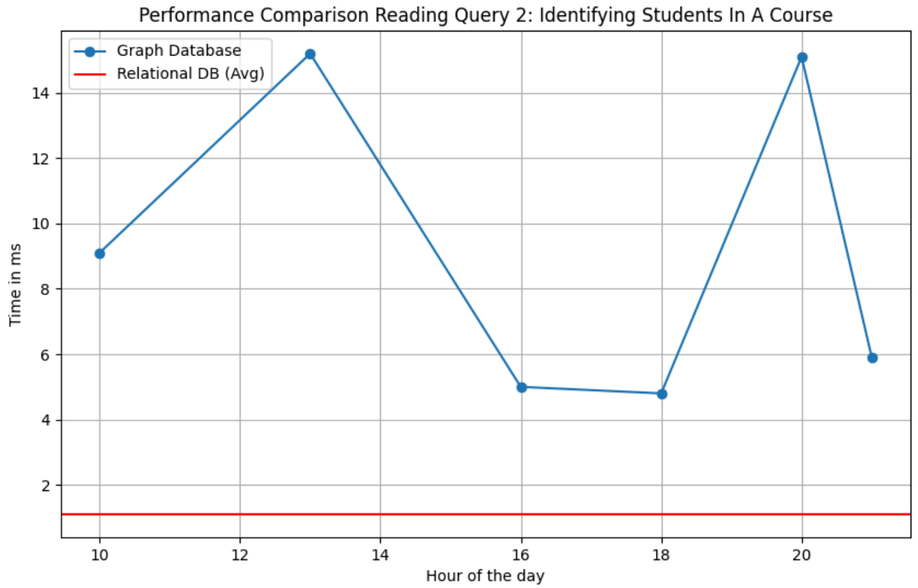
Risk Students



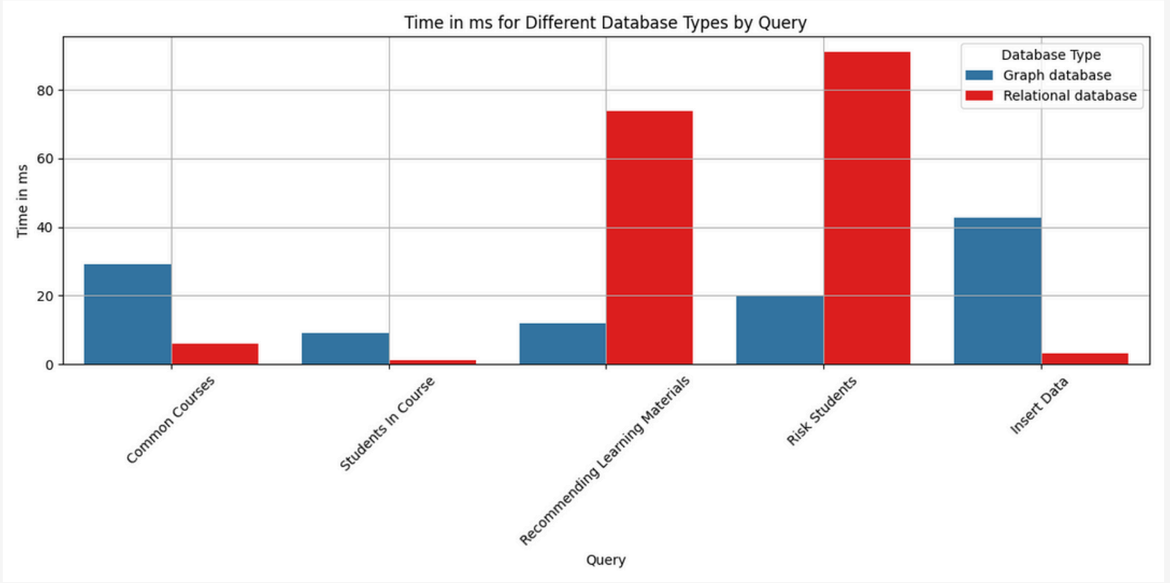
Common Courses



Results



Students In A Course



Performance Comparison

- **Query Efficiency:**
  - graph database excels in complex queries (recommendations and risk identification)
  - relational database outperforms in simple queries (data insertion, querying students)

Discussion

Limitations:

- Tests on a limited dataset.
- Neo4j AuraDB Free limited to 200k nodes/400k relationships.
- Online hosting may introduce latency.
- Neo4j's initial queries slower due to disk access before caching.

Insert Performance:

- Graph DB response times: 14.1 ms - 70 ms; prioritizes efficient data analysis.
- Relational DB insert times increase with table size.

Graph DB Advantages:

- .Efficient for suggesting study materials and identifying at-risk students.
- Handles complex relationships and deep queries efficiently.

Relational DB Advantages:

- Faster for simple queries with fewer relationships.
- Optimized for direct access and join operations.

Theoretical Outlook

01

Vendor Lock-In:

- Neo4j supports multiple programming languages
- Cypher, is widely used
- Migration Tools, Data Import Techniques

02

Cluster Arrangement:

- Neo4j uses a master-slave setup ensuring high availability and minimal downtime.

03

Scalability:

- Neo4j's Autonomous Clustering supports horizontal scaling for higher throughput.
- Sharding with Neo4j Fabric: Enables storage of large graphs in smaller independent graphs for easy scale-out. Maintains performance regardless of graph size, dependent on query complexity.

04

CAP Theorem:

- Neo4j prioritizes consistency and partition tolerance over availability.

05

Schema Changes:

- Adding new nodes and relationships usually doesn't affect existing queries.
- Simplifies changing existing relationships or node structures compared to other databases.

Conclusion

Neo4j Strengths

- ideal for managing complex relationships, real-time analytics and personalized learning in educational settings

Neo4j Weakness

- **Data Insertion**
  - less efficient than PostgreSQL for data insertion tasks (at the time)

Optimization Recommendations

- data inserts via batch processing during low activity periods
- use of tools like LOAD_csv for effective bulk data import
- consideration of upgrading to paid version